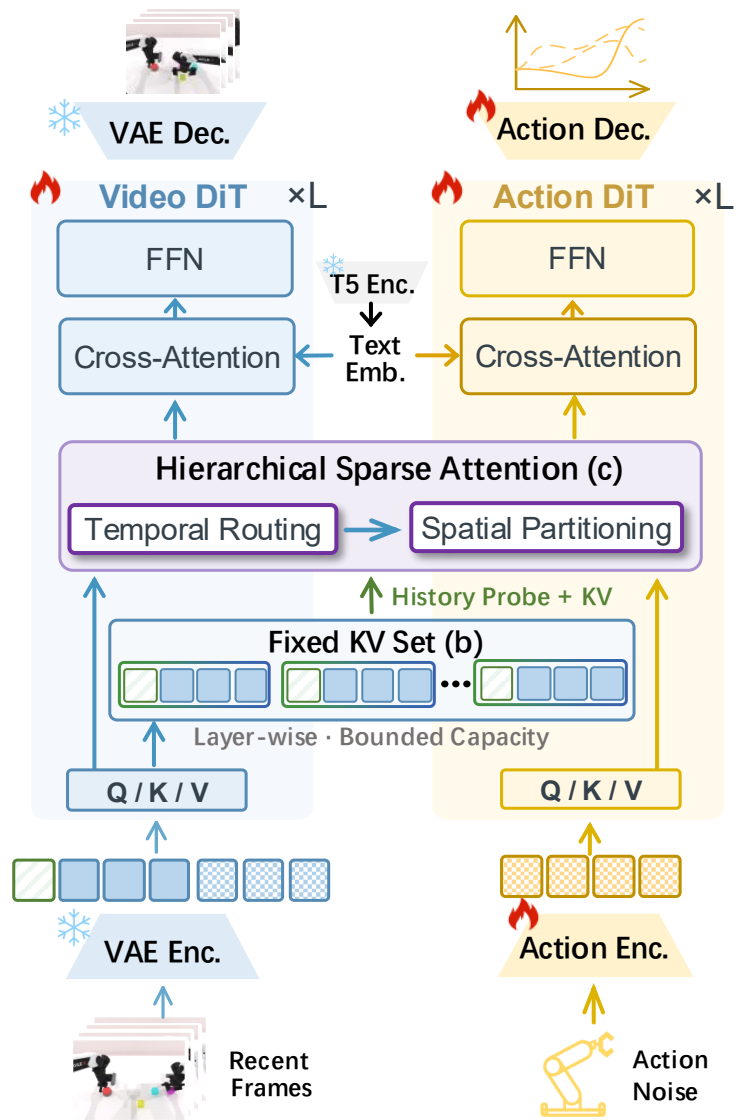
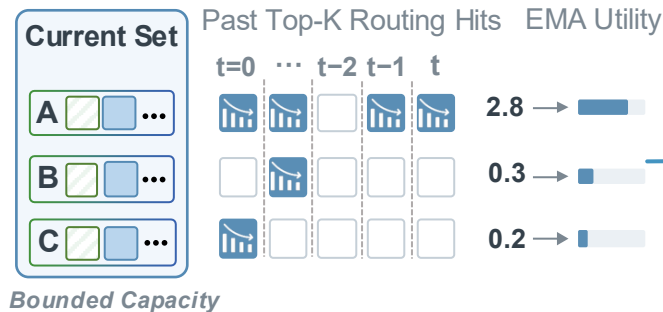


(a) SHARP-WAM Architecture

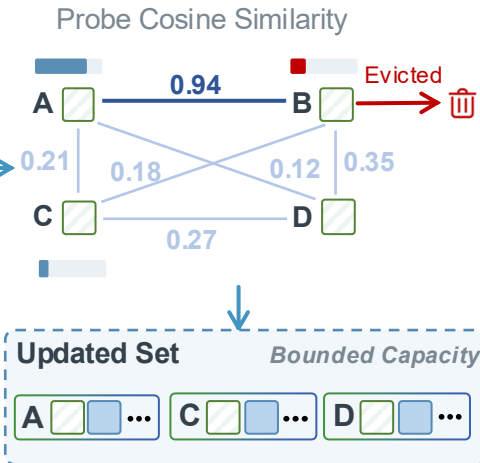


(b) Fixed KV Set and EMA-utility Eviction

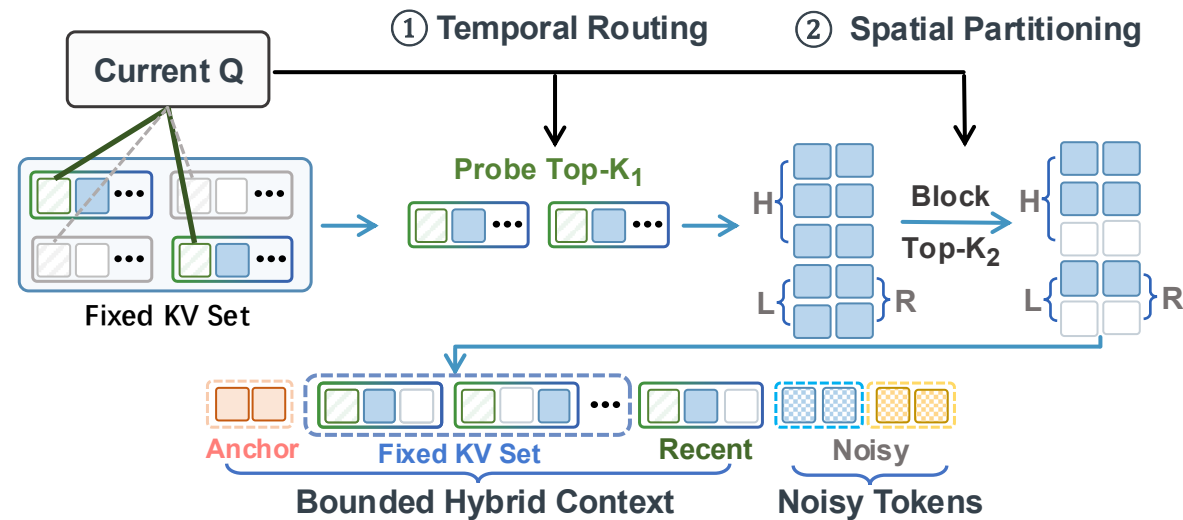
① Time-Decayed EMA Utility



② Compare & Evict



(c) Hierarchical Sparse Attention



Clean Video Token
 Noisy Video/Action Token
 Probe Token
 NOT Routed
 Anchor token
 — Top-K Routed
 --- Top-K NOT Routed